



Use of Cell Phones in Near Surface Seismology

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Abstract. The experiment aimed to test the possibility of using phones to measure signals and waves used in near-surface seismology instead of specialized geophones. Yeouido Han River Park was selected based on the presence of a grassland, high cultural sounds, a subway line passing underground, and deposited quaternary alluvium. Using Sensor Log, an app that the acceleration of the phone using an accelerometer, cultural noise experiment and hammer energy source experiment were conducted. In the cultural noise experiment, data was collected to explore the possibility of phones being able to collect cultural seismic waves; in the hammer energy source experiment, data was collected to explore the possibility of phones being able to collect seismic waves from a particular energy source. MATLAB was used to analyze the data and create a visual representation. The results revealed that it is possible to use phones as a replacement for geophones, allowing the analysis of surface waves possible without specialized equipment in wide arrays of situations.

Keywords: Accelerometer · Cultural Noise · Near Surface Seismology · Phones · Seoul

1 Introduction

This study explores the possibility of replacing Multichannel Analysis of Surface Waves (MASW) with phones and its accelerometer board.

1.1 Near Surface Seismology

Near surface seismology is a technique that uses seismic data to characterize the mechanical properties of the near surface (0–30 m). There are several techniques, such as MASW [5]. MASW uses the phase velocity of surface wave arrivals at different frequencies to recover shear velocity versus depth. With 3-component seismometers, a technique known as the H/V method has been used to estimate the depth to the basement [2].

Typically, a built-for-purpose acquisition system including an active source such as a weight drop or hammer is used along with geophones for each of these methods. In this study, we examine the possibility of using the 3-component accelerometers with cell phones as a replacement for conventional geophones. With such a system, investigators could collect data at a fraction of the cost.

1.2 Phone Accelerometer

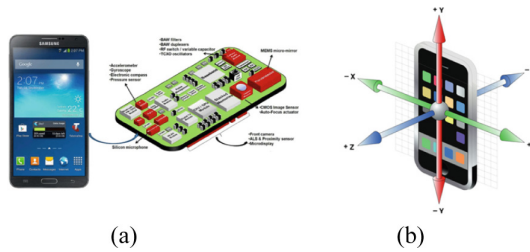


Fig. 1. Phone Accelerometer Board (a) <https://dealna.com/Article/Post/313/How-the-Accelerometer-in-Your-Phone-Could-Help-Identify-You> (b) <https://turbofuture.com/cell-phones/10-Best-Accelerometer-Apps>

As shown in Fig. 1, a phone accelerometer is a built-in sensor that measures the force of acceleration caused by movements or gravitational force.

2 Data Acquisition Method

2.1 Selection of Location

Area selection was based on the presence of a grassland, high cultural sounds, a subway line passing underground, and deposited quaternary alluvium. Proximity to the river ensured that quaternary alluvium was deposited in the area. As such, the area along the Han River was examined to find experimental locations. Figure 2 indicates possible areas along the Han River [3].

Along the Han River, parks, which had grassland, were established. Among all the Han River parks, parks without subway lines passing underneath and parks with low cultural sound were eliminated.

After this process, Yeouido Han River Park was selected as the experiment site. As Fig. 3 indicates, the purple subway line passes through the grassland area of the park. Moreover, adjacent to the park is a four-lane road that can provide high cultural sounds.

2.2 Sensor Log APP

Data collection was executed through an app called Sensor Log. To measure using Sensor Log, the phone should be placed on a flat land with the screen facing up without any phone case or other accessories. As shown in Fig. 4, Sensor Log measures the acceleration of the phone in three-axis. For each measurement, the acceleration is recorded at a logging rate of 100 Hz and is converted into a CSV file. CSV file contains information of the time stamp and acceleration in each axis. Later, the CSV file is analyzed using MATLAB.

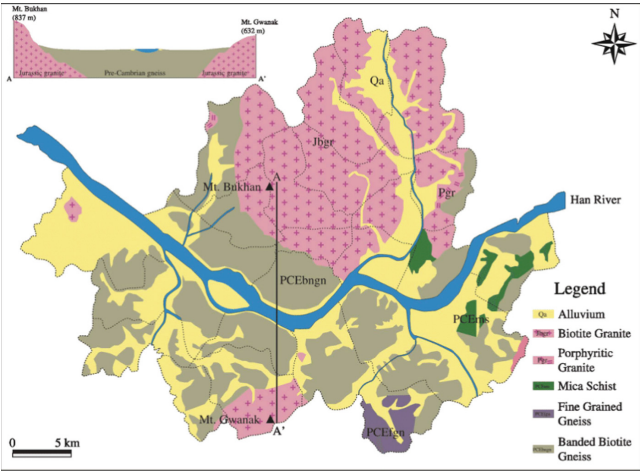


Fig. 2. Map of Seoul based on Geologic Features https://www.researchgate.net/figure/Simplified-geological-map-of-Seoul-Modified-from-Kim-et-al-2016_fig8_32204463

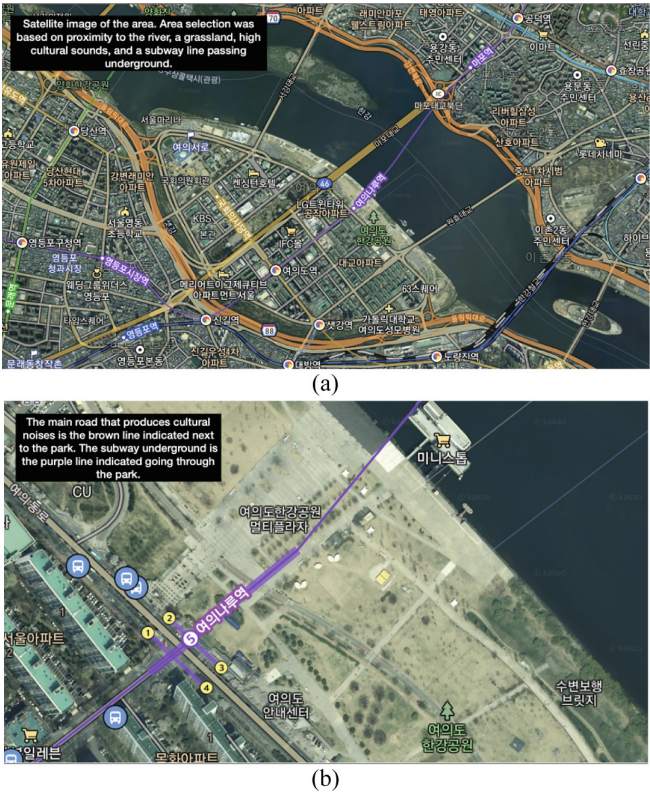


Fig. 3. Map of Yeouido Han River Park

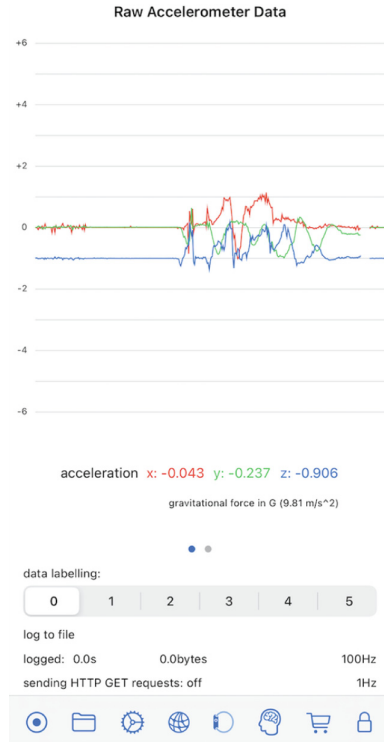


Fig. 4. Sensor Log Screen Shot

2.3 Phone Set-Up

The first set of experiments focused on determining what type of phone set-up would be most effective in capturing seismic waves. The experiment has three set-ups: phone without any supplements, phone pressed with a heavy book, and phone attached to a plastic stake. A simple experiment showed that the plastic stake was the most effective method to measure seismic waves.

To effectively insert the stake into the grassland, the stake was first hammered into the ground as the phone is set up in Fig. 5. When the stake was hammered enough so that the head of the stake is at ground level, double-sided tape and strong glue were used to attach the phone to the head of the stake.

Then, another experiment was conducted to determine the time of the day that the phone can most effectively collect data. Using a phone attached to a plastic stake, the measurement was taken during the early morning (5:00–7:00 A.M.), mid-day (11:00 A.M.–1:00 P.M.), and late night (10:00 P.M.–12:00 A.M.). The time range was restricted by the operation time of the subway. The results showed that measurements taken during the early morning contained less ground-level cultural noise and clearer seismic waves.

Lastly, among different versions of iPhones (X, 11, and 12), iPhone 12 was the most efficient in collecting data.

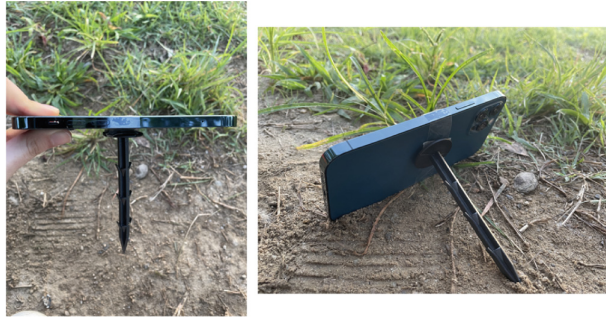


Fig. 5. Image of a Phone Attached to a Plastic Stake

2.4 Cultural Noise Experiment

Cultural Noise Experiment was conducted to collect cultural seismic waves, such as the waves from the subway line, using phones.

Three iPhone 12s were used to collect data on August 7th, 2022, between 5:40 A.M. and 7:00 A.M., with each measurement being 15 min long. Experiment time was restricted due to starting time of subway operation and increasing ground-level cultural noise. Thus, the experiment was conducted for two consecutive days to measure 14 different locations as in Fig. 6.

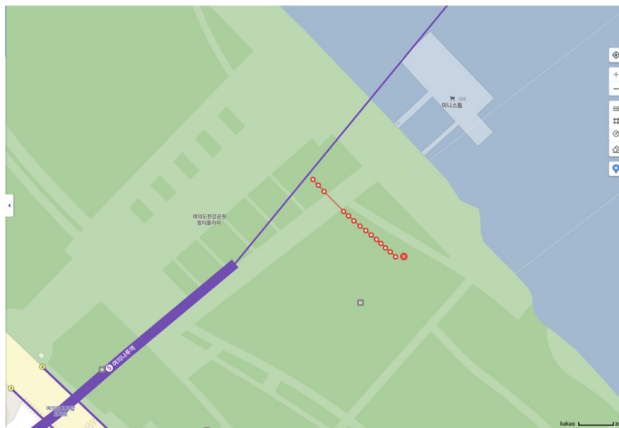


Fig. 6. Cultural Noise Experiment Phone Location on Red Dots

The phones are placed in a line that is perpendicular to the subway line. Also, the phone was placed so that the configuration of the phone itself is parallel with the line created by the phones, and the phones are directed towards the subway line. The reference phone is placed at a three-meter distance from the subway line and is fixed and not moved for all measurements. As shown in Fig. 7, starting from the reference phone, phones are placed in three-meter intervals—if there were an obstacle, the phone was placed after

the obstacle. Due to limitations in the number of phones that could be utilized, for every measurement, two iPhone 12s (excluding one used as a reference phone) were moved according to the interval.

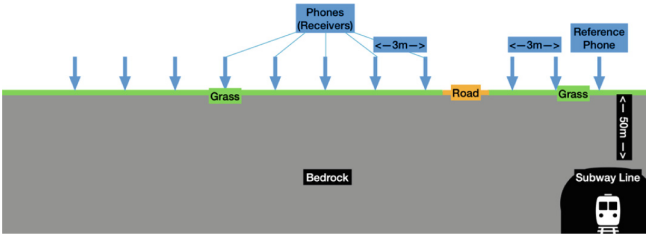


Fig. 7. Cultural Noise Experiment Set-Up

2.5 Hammer Energy Source Experiment

Hammer Energy Source Experiment was conducted to collect seismic waves from an energy source, such as the waves from hammering a steel plate, using a phone.

Again, three iPhone 12s were used to collect data on August 28th, 2022, between 5:40 A.M. and 7:00 A.M. For each phone location, two measurements were made: five minutes of no hammering and 40 s of hammering.

For two minutes of hammering, the steel plate used was 20 cm $\times$ 20 cm $\times$ 2 cm large, as shown in Fig. 8. The steel plate was placed on a flat grassland, and for 40 s, the steel plate was hit with a hammer with full power every 10 s.

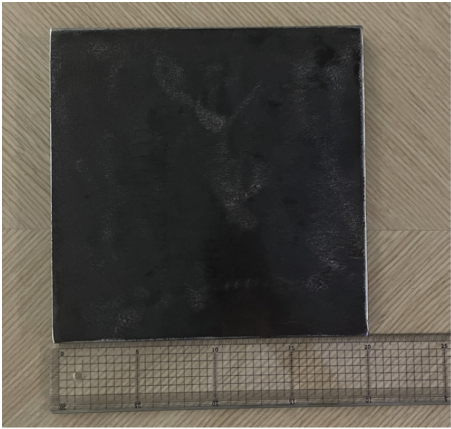


Fig. 8. Steel Plate

The phones were placed so that a linear line, including the hammer source, is created. The reference phone was placed at a two-meter distance from the energy source and was

not moved throughout the whole experiment. The phone was set up as shown in Fig. 9. From the reference phone location, phones, two iPhone 12s, were moved according to the two-meter intervals for each measurement.

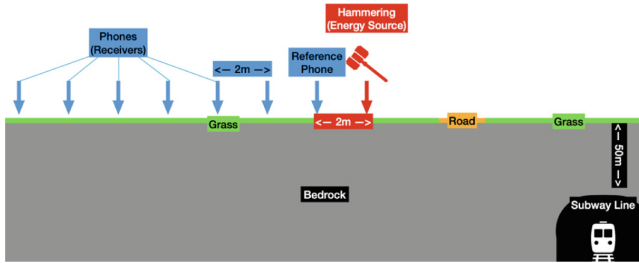


Fig. 9. Hammer Energy Source Experiment

3 Experimental Results

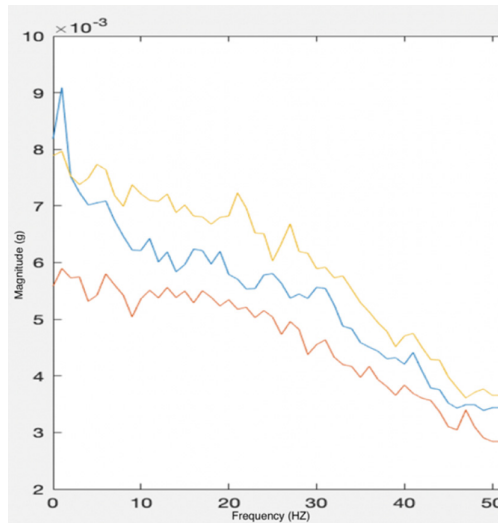


Fig. 10. Cultural Noise in High Traffic Time Results (Z-Component: Dark Orange, X-Component: Blue, Y-Component: Yellow)

Figure 10 shows the Fourier magnitude spectrum for a receiver 12 m from the reference phone computed from approximately x seconds of data. The data were band-passed from 5–45 Hz prior to computing the magnitude spectrum. The average spectral magnitude is around 5×10^{-3} g, which is an order of magnitude above the -5×10^{-4} g self-noise level of the cell phone accelerometers and indicates that each component is measuring cultural noise signals. The y component has a large spectral peak between 30 and 35 Hz, which is similar to the spectral peak observed in another study measuring subway-generated noise with a conventional geophone [4].

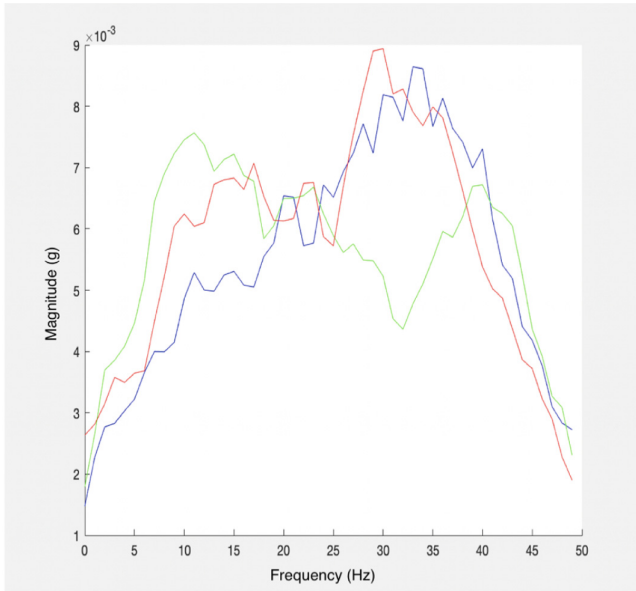


Fig. 11. Hammer Energy Source Results (Z-Component: Blue, X-Component: Red, Y-Component: Green)

Figure 11 shows the Fourier magnitude spectrum for a similar receiver location from the hammer-on-plate data. The data were band-passed from 5–45 Hz prior to computing the magnitude spectrum. Note that the y component is smallest in this data and that the frequency spectrum is decidedly peaked at much higher frequencies than the cultural noise data. Hammer on plate data is known to be weak at low frequencies, unlike cultural noise recordings. Moreover, the predominant signals are expected in the vertical plane from a hammer source.

4 Discussion and Conclusion

The experimental results indicate that it is possible to replace phones with specialized devices to measure signals and waves of use in near-surface seismology.

However, some caveats exist in this method. Since the phones were not equipped with anything than timing the measurements with a separate device, the measurement could not be turned on and off at the exact moment. Furthermore, because the measurements had to be manually turned on and off, the first and last few seconds of the measurements had to be trimmed out during the analysis process. Further developments of this technique could create a trigger switch that would accommodate the timing of the measurements.

This technique of replacing geophones with commonly used cell phones allows seismic research to be done without any specialized equipment and in a variety of settings. When doing archaeological research, near-surface seismology research would be effective in determining the presence of artifacts or human remains underground. In those situations, without using specialized equipment for geologic research and by collecting

seismic waves and signals using phones, near-surface seismology could be effectively utilized to aid the research. Furthermore, in civil engineering and urban architecture, near-surface seismology could be used to determine the presence of sewerage systems and other underground structures. This technique could significantly reduce the cost and complicatedness of using near-surface seismology analysis.

Especially in South Korea, there are multiple instances where construction is delayed for several years due to archaeological remains discovered under the site [1]. Using near-surface seismology and analyzing underground substances before initiating the actual construction could lead to huge economic and social benefits. Furthermore, there is historical significance as valuable artifacts and remains could be excavated using proper and careful methods rather than being discovered during urban construction.

References

1. Bandun, R.: [Cityscapes] Exploring beneath Gwanghwamun's Surface. Koreatimes. 27 Apr 2021. www.koreatimes.co.kr/www/nation/2021/05/177_307682.html
2. Hong, T.-K., et al.: Roles of subway speed and configuration on subway-induced seismic noises in an urban region. *J. Appl. Geophys.* 13 May 2022, www.sciencedirect.com/science/article/abs/pii/S0926985122001392?fr=RR-2&ref=pdf_download&rr=7632b5dbfb669325
3. Lee, J.-Y., Kwon, K.D., Raza, M.: Current water uses, related risks, and management options for Seoul megacity, Korea. *Environ. Earth Sci.* **77**(1), 1–20 (2017). <https://doi.org/10.1007/s12665-017-7192-6>
4. Psarropoulos, P.: Impact of Tunnels and Underground Spaces on the Seismic Response of Overlying Structures. IntechOpen, IntechOpen, 3 December 2019. www.intechopen.com/chapters/69298
5. Jumrik, T., Dey, A.: A review of active and passive MASW techniques. In: National Workshop Engineering Geophysics for Civil Engineering and Geo-Hazards (EGCEG) (2012)